

SHIRAN DUDY

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PROFESSIONAL SUMMARY

Research Scientist with 10+ years of experience in NLP, machine learning, and responsible AI. Deep expertise in evaluating AI systems for fairness, cultural representation, and bias, with a strong record of translating technical findings into actionable guidance for cross-functional stakeholders. Passionate about advancing civil liberties through evidence-based, equity-centered technology research and deployment.

CORE COMPETENCIES

NLP & Language Modeling • Algorithmic Fairness & Bias Evaluation • AI Governance & Responsible Deployment • Generative AI Systems (LLMs/SLMs) • Python (PyTorch, HuggingFace, LangChain, NLTK) • Participatory & Mixed-Methods Research • Cross-Functional Communication • Cloud Services (AWS EC2, Docker) • Policy & Regulatory Text Analysis • Stakeholder Engagement

PROFESSIONAL EXPERIENCE

Research Scientist, Responsible AI | Northeastern University | Boston, MA

2023 – Present

- Lead technical responsible AI consulting, guiding organizations in developing and deploying AI systems that meet standards of fairness, robustness, and accountability — directly analogous to the ACLU's AI governance goals.
- Evaluate proposed AI use cases for cultural representation, bias, and equitable impact across diverse populations, producing actionable recommendations for technical and non-technical stakeholders.
- Design and conduct participatory research centering affected communities' voices in AI deployment decisions, including co-design sessions at FAccT 2025.
- Prepare AI modules and curricula on responsible AI for professionals and organizations; communicate complex NLP concepts to diverse audiences.
- Mentor research teams (co-ops, interns, postdocs) on responsible AI projects.
- PI on "Seeing the world through LLM-colored glasses" project (\$60k, 2024), investigating bias in AI information systems.

Research Associate, Human-Computer Interaction | University of Colorado Boulder | Boulder, CO

2021 – 2023

- Researched and developed conversational AI systems designed to promote equitable and respectful collaboration in educational settings, with attention to fairness across diverse demographics.
- Led interruption detection and equity-oriented intent schema projects for the Institute for Student-AI Teaming (iSAT), applying NLP methods to real-world interaction data.
- Contributed to speaker diarization research analyzing equity in classroom group discussions (EDM 2024, Best Student Short Paper Award).
- Published on personalization and relevance in NLG (EMNLP 2021), exploring how language models serve diverse user needs.

Ph.D. Researcher, Computer Science (NLP & Accessibility) | Oregon Health & Science University | Portland, OR

2013 – 2020

- Developed and evaluated language models (FST, Transformers) for brain-computer interface systems, investigating model biases across symbol sets to ensure equitable access for users of varying literacy levels.

- Conducted statistical analyses on text data and machine learning evaluation methods; published on evaluation metrics for NLG (EMNLP 2020, 2nd Best Paper Award).

EDUCATION

Ph.D., Computer Science | Oregon Health & Science University | Portland, OR 2013 – 2020
Focus: NLP, Machine Learning, Accessibility, and Fairness

B.Sc., Biomedical Engineering | Ben Gurion University | Israel 2008 – 2012
Focus: Speech Signal Processing, Accessibility

SELECTED PUBLICATIONS

- “Taking Stock at FAccT”: Using Participatory Design to Co-Create a Vision for the Fairness, Accountability and Transparency Community”-- FAccT 2026
- “Unequal Opportunities: Examining the Bias in Geographical Recommendations by Large Language Models” — IUI 2025 (Honorable Mention Award)
- “Analyzing Cultural Representations of Emotions in LLMs through Mixed Emotion Survey” — ACII 2024 (Best Paper Award)
- “Bias in Retrieval Systems” — book chapter, ACM 2024
- “Expansive Participatory AI: Supporting Dreaming within Inequitable Institutions” — NeurIPS HCAI Workshop 2022
- “Refocusing on Relevance: Personalization in NLG” — EMNLP 2021
- “Are Some Words Worth More Than Others” — EMNLP Eval4NLP 2020 (2nd Best Paper Award)

POLICY, GOVERNANCE & COMMUNITY ENGAGEMENT (volunteering)

- Tech Policy Tracker: Lead a volunteer team developing a tool to make technology policy accessible to everyday people; collaborate with the Integrity Institute on outreach and funding.
- Policy tracker for civil society advocacy groups: co-designing a tracker with (and for) the Phone-free school movement in collaboration with Boston University BU!Spark team.
- IAAA Member: Participate in establishing industry standards for AI product certification and auditor training (International Association for AI Auditors).
- Advisory Council Member: Massachusetts Science and Technology Policy Fellowship (MASTPF), bridging science and public service in the State House.
- Organize NLPerspectives workshops (LREC 2026, EMNLP 2025, LREC-COLING 2024, CEUR-WS 2023) and co-design sessions at FAccT 2025, centering diverse perspectives in NLP research.
- Elected as one of 100 Brilliant Women in AI Ethics, 2023.

TECHNICAL SKILLS

Languages: Python, R, C++, Bash/Shell | ML/NLP: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, NLTK | Infrastructure: Docker, SLURM, AWS EC2, Git | Research Methods: Experiment design, IRB-led human subjects research, mixed methods (semi-structured interviews, surveys, quantitative analysis)